Projected Local Projections

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Abstract

This paper introduces a nonparametric estimator for functions defined over discrete, naturally ordered supports and applies it to impulse response function estimation within the local projections (LP) framework. The estimator allows for the possibility that the function admits a lower-dimensional representation by projecting it onto a basis of piecewise continuous functions. To account for uncertainty about the appropriate dimensionality in estimation and inference, the approach employs model averaging across specifications with different dimensions, yielding the Averaged Projected Local Projections (APLP) estimator. Extensive simulations show that APLP has lower variance than the standard LP estimator while introducing little bias, and achieves similar coverage with shorter confidence intervals; the median APLP interval is about 20% shorter. Two applications demonstrate that APLP improves interpretability by smoothing estimates across horizons and tightening confidence bands.

Keywords: nonparametric regression, jackknife model averaging, forward guidance, yield curve, risk premia

JEL classification codes: C14, C22, C32

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1 Introduction

Many empirical economic questions involve estimating functions defined over discrete, naturally ordered supports: $\beta : \{H_{\min}, \ldots, H_{\max}\} \to \mathbb{R}$. In macroeconomic applications, a key example is impulse response function (IRF) estimation, which measures the effect of a shock x_t on an outcome y_{t+h} at different horizons h:

$$\beta(h) = \mathbb{E}\left[\frac{\partial y_{t+h}}{\partial x_t}\right], \quad h \in \{H_{\min}, \ldots, H_{\max}\}. \tag{1}$$

IRFs are central inputs to policy-relevant statistics such as the fiscal multiplier. As another example, in life-cycle analysis, h indexes age and the conditional expectation of an outcome, such as earnings, can be written as:

$$\beta(h) = E(Y_i \mid X_i = h), \quad h \in \{H_{\min}, \ldots, H_{\max}\}. \tag{2}$$

A flexible and simple nonparametric estimator for β is, in the case of Equation (2), to compute conditional averages for each age group separately. For Equation (1), one can project y_{t+h} on x_t for every horizon h, which is the well-known and popular Local Projection (Jordà, 2005) estimator. From a function approximation view, this is as if we project β on the base of indicator functions: $\beta(h) = \sum_{k=H_{\min}}^{H_{\max}} I(k = h)\beta_h$, where β_h is the h^{th} element of the finite-dimensional vector $\boldsymbol{\beta} = [\beta(H_{\min}), \ldots, \beta(H_{\max})]$. While flexible, this comes at a cost: there are as many free parameters as the cardinality of the support, denote $H = (H_{\max} - H_{\min} + 1)$, leading to high variance.

This paper proposes an alternative nonparametric estimator for impulse response functions and other functions with discrete ordered support, by conjecturing that $\boldsymbol{\beta}$ may (but does not have to) admit a lower-dimensional representation:

$$\boldsymbol{\beta}^* = P_k \boldsymbol{\beta}, \tag{3}$$

where P_k is a projection matrix of rank $k \le H$ that restricts β to be a piecewise continuous function. When $k < H$, this results in dimension reduction.

Evidently, there are many possible bases one can project $\boldsymbol{\beta}$ onto, both linear and nonlinear. Focusing on impulse response function estimation, examples include the base of indicator functions just mentioned, Gaussian basis functions (Barnichon and

Matthes, 2018), or B-splines (Barnichon and Brownlees, 2019). Similarly, economic models such as Dynamic Stochastic General Equilibrium (DSGE) frameworks and vector autoregressions (VAR) imply certain functional forms for IRFs, but their inherent parametric nature raises concerns about model misspecification. Piecewise continuous functions are attractive because: (i) they are more efficient than indicator function bases, (ii) unlike Gaussian basis functions, they can be mapped into the class of linear estimators, and (iii) they are more parsimonious than B-splines. Their parsimoniousness hinges on being able to estimate the segment locations of the piecewise continuous functions. Hence, part of our contribution is a computational algorithm that asymptotically recovers optimal segment locations, resulting in a Projected Local Projection (PLP).

The rank of the projection matrix in Equation (3), corresponding to the number of free parameters k, has to be chosen by the econometrician. A naive approach would select k using model selection criteria, but this introduces post-selection bias. Instead, to account for uncertainty in k, we build on Jackknife model averaging methods (Hansen and Racine, 2012; Zhang and Liu, 2019; Liu, 2015), resulting in our Averaged Projected Local Projection (APLP) estimator. Note that since segment locations estimation is super consistent, it suffices asymptotically to account only for rank uncertainty. The APLP estimator is computationally efficient, delivers mean squared error improvements relative to the unrestricted local projection estimator, is consistent, and enables valid inference.

We illustrate the performance of the APLP estimator using two simulation settings and two empirical applications. The first simulation focuses on ARMA(p, q) processes with impulse response shapes often encountered in macroeconomics, comparing our nonparametric estimator APLP to standard local projections (LP), bias-corrected local projections (BCLP, Herbst and Johannsen, 2024), and smooth local projections (SLP, Barnichon and Brownlees, 2019). Beyond bias and variance, we evaluate confidence interval length and coverage. For sample sizes $T = 100$ and $T = 200$, SLP severely undercovers, due to both substantial bias, especially at short horizons, and their proposed confidence intervals being too narrow. In contrast, under the asymptotic framework of Zhang and Liu (2019), APLP achieves comparable coverage to LP and BCLP with shorter intervals (15-30% shorter), or, under the alternative asymptotic framework of Liu (2015), can improve upon LP coverage with the same confidence interval length. Regarding simultaneous confidence bands (as in Montiel Olea and Plagborg-Møller, 2019), conservative Bonferroni simultaneous confidence bands show SLP coverage dropping as low as 25%, whereas APLP outperforms LP and BCLP in both coverage and confidence band width.

To complement this stylized simulation, we also evaluate performance in the large-scale simulation design of Li et al. (2024), which enables comparison of local projection methods across thousands of data-generating processes representative of the U.S. economy. While Li et al. (2024) focuses on point estimator properties, we extend the analysis to include confidence band width and coverage. The results confirm that our earlier findings generalize: APLP achieves similar coverage to standard LPs but with narrower bands, and consistently outperforms both LP and BCLP in terms of mean squared error. Although SLP exhibits lower variance, it introduces substantially more bias, and its confidence bands severely undercover.

Finally, we demonstrate the method in two empirical applications. The first revisits the effect of monetary policy on risk premia, following the analysis in Kekre and Lenel (2022) and using the monetary policy shocks from Gertler and Karadi (2015). We estimate impulse responses using standard local projections, our APLP method, and smooth local projections. APLP yields smoother point estimates than LP while closely tracking its trajectory. Moreover, APLP produces narrower confidence bands, resulting in impulse responses that are statistically significant at more consecutive horizons than those from LP. SLP also produces smooth estimates and, at times, narrower confidence bands; however, its shrinkage toward zero can result in none of the horizons being significantly different from zero.

In the second application, we examine the effects of the federal funds rate (FFR) and forward guidance (FG) shocks, as identified in Swanson (2021), on the yield curve. We show that FFR and FG shocks have markedly different effects: FG shocks affect the yield curve with a delay, peaking around 4–5 years after the shock, while FFR shocks operate more quickly and peak at approximately 1.5 years. When high-frequency (weekly) data are available, APLP and LP produce similar estimates. However, in lower-frequency settings (monthly or quarterly), APLP delivers smoother point estimates and tighter confidence bands, with more periods significantly different from zero than LP.

Related Literature Our proposed computational algorithm for recovering optimal segments in piecewise continuous functions is novel, and relates closely to the dynamic programming algorithms used in breakpoint detection (Bai and Perron, 2003), slope change detection (Fearnhead et al., 2019), and others.

While our proposed nonparametric estimation method has broader applicability, as highlighted in the introduction, our main application focuses on impulse response function estimation. We therefore emphasize its advantages relative to the existing literature in this area. Beyond the papers already discussed, related contributions include Ho et al. (2024), who use prediction pools for model averaging between local projections and DSGE models or VARs, and Ferreira et al. (2025), who employ quasi-Bayesian estimation of local projections with a prior obtained from a VAR to shrink toward VAR estimates.

Relative to these approaches, our method offers two important conceptual advantages: (i) it retains the semi-parametric nature of local projections, and (ii) it delivers valid inference. Regarding (i), in contrast to Ho et al. (2024) and Ferreira et al. (2025), we do not shrink toward parametric models such as VARs or DSGEs, thereby preserving the semi-parametric structure that is a key appeal of local projections. Regarding (ii), while Ho et al. (2024) also employ model averaging, their inference does not account for model weight uncertainty, which tends to lead to undercoverage. This is particularly problematic since local projections already tend to undercover (e.g., as documented in Herbst and Johannsen, 2024). By contrast, our inference explicitly accounts for model weight uncertainty and therefore remains valid. Similarly, Barnichon and Brownlees (2019) provide *heuristic* confidence intervals, which also tend to undercover.

2 Piecewise Function Approximation

This section introduces the general layout for piecewise approximations by low-order polynomials with continuity imposed between breakpoints. The approach delivers a flexible fit with a small number of parameters and produces smoother impulse responses than unconstrained horizon-by-horizon OLS.

Unrestricted Model Consider the stacked local projection system:

$$y_{t+h} \;=\; \beta(h)\, x_t + \varepsilon_{t+h}, \qquad h = 0, 1, \ldots, H, \;\; t = 1, \ldots, T-h, \tag{4}$$

where $x_t \in \mathbb{R}$ is observed at time t, $y_{t+h} \in \mathbb{R}$ is the h-step response, and $\beta(h)$ is the impulse response at horizon h. Unconstrained OLS minimizes

$$\min_{\boldsymbol{\beta}} \sum_{h=0}^{H} \sum_{t=1}^{T-h} \big(y_{t+h} - \beta_h x_t\big)^2, \tag{5}$$

and is separable across h, yielding $\hat{\beta}_h$ for $h = 0, \ldots, H$.

Restricted Models Suppose, instead, structure is imposed on $\beta(\cdot)$. Let the grid of horizons $\{0, 1, \ldots, H\}$ be partitioned by breakpoints $\boldsymbol{\tau} = (\tau_0, \tau_1, \ldots, \tau_K)$ with $0 = \tau_0 < \tau_1 < \cdots < \tau_{K-1} < \tau_K = H$. Define the segments

$$S_1 = \{0, \ldots, \tau_1\}, \qquad S_k = \{\tau_{k-1} + 1, \ldots, \tau_k\} \text{ for } k \geq 2.$$

On segment k, let the local coordinate be $s = h - \tau_{k-1}$ and posit a segment-specific map $f_k(s; \theta_k)$. Some examples are:

$$\begin{aligned}
\text{intercept only: } & f_k(s;\theta_k) = a_k, \text{ where } \theta_k = a_k, \\
\text{linear: } & f_k(s;\theta_k) = a_k + b_k s, \text{ where } \theta_k = (a_k, b_k), \\
\text{quadratic: } & f_k(s;\theta_k) = a_k + b_k s + c_k s^2, \text{ where } \theta_k = (a_k, b_k, c_k).
\end{aligned}$$

Let $\boldsymbol{\theta} = (\boldsymbol{\tau}, \theta_1, \ldots, \theta_K)$ collect all parameters. The impulse response can then be written as

$$\beta(h; \boldsymbol{\theta}) \;=\; \sum_{k=1}^{K} \mathbb{I}\{h \in S_k\}\, f_k\big(h - \tau_{k-1};\, \theta_k\big). \tag{6}$$

Note, the unrestricted model is an *intercept only* model with as many segments as horizons.

Continuity Constraints In addition to positing a functional form for each segment, we may impose restrictions that produce *smooth* impulse responses. Two natural smoothness restrictions are level continuity C^0 and slope continuity C^1 at breakpoints, defined as:

$$\begin{aligned}
C^0 :& f_{k-1}(\tau_{k-1}) = f_k(\tau_{k-1}),\\
C^1 :& f_{k-1}(\tau_{k-1}+1) - f_{k-1}(\tau_{k-1}) = f_k(\tau_{k-1}+1) - f_k(\tau_{k-1}),
\end{aligned}$$

where the former prevents *jumps* and the latter prevents *kinks*; together they reduce the jaggedness typically found in unrestricted local projections. The number of restrictions that can be imposed depends on the number of free parameters in each segment. Conditional on breakpoints $\boldsymbol{\tau}$, these restrictions give rise to a constraint set $C(\boldsymbol{\tau})$ for the segment coefficients that satisfy the constraints.

Objective Given K, we estimate the integer breakpoints $\boldsymbol{\tau} = (\tau_0, \tau_1, \ldots, \tau_K)$ with $0 = \tau_0 < \tau_1 < \cdots < \tau_K = H$ and the segment coefficients $\boldsymbol{b}$ (intercepts and slopes) by

$$\min_{\boldsymbol{\tau}, \{\theta_k\} \in C(\boldsymbol{\tau})} \sum_{k=1}^{K} \sum_{h=\tau_{k-1}+1}^{\tau_k} \sum_{t=1}^{T-h} \left\{ y_{t+h} - \beta(h; \boldsymbol{\theta}, \boldsymbol{\tau})\, x_t \right\}^2, \tag{7}$$

optionally, subject to a minimum segment length constraint $\tau_k - \tau_{k-1} \geq \ell$ (with $\ell \geq 1$). For fixed $\boldsymbol{\tau}$, (7) is a linear least squares problem in $\boldsymbol{\theta}$. The presence of free breakpoints and continuity constraints (which depend on the breakpoint locations) makes this a challenging optimization problem, which we address in the next sections.

Dynamic Programming and Computational Efficiency The problem of choosing optimal breakpoints for least-squares problems without continuity constraints is known as the *segmented least-squares* problem, and a dynamic-programming algorithm can be found in various textbooks (see, e.g., Kleinberg and Tardos (2006)). Unlike that setting, here the intercept (and possibly the slope) on each segment is the value inherited from the previous segment's endpoint, which makes naive precomputation of all segment costs impossible. We therefore cast the problem in a dynamic-programming formulation with a state that includes the segment-endpoint value. Let $C(k, j)$ denote the minimal residual sum of squares achievable with k segments ending at horizon j, and let $\gamma_{0,k-1}(i)$ denote the inherited value and $\gamma_{1,k-1}(i)$ denote the inherited slope at the start of the kth segment when the kth segment begins at horizon i. A Bellman recursion over admissible predecessors $i < j$

chooses the optimal split i between segments $k-1$ and k This reduces the search from the combinatorial number of partitions $\binom{H-1}{K-1}$ to $O(KH^2)$ transitions. For example, when $H = 48$ and $K = 12$, the number of partitions is $\binom{47}{11} \approx 1.74 \times 10^{10}$, so simply enumerating all possible breaks is infeasible, whereas the dynamic program is practical and the solution can be found in a fraction of a second.

Algorithm Here we outline the algorithm used to find the optimal breakpoints for the piecewise quadratic form of $\beta(h)$ with C^0 and C^1 continuity imposed. We summarize the entire procedure compactly in Algorithm 1.

Let $\{(y_t, x_t)\}_{t=1}^T$ be observed. For each $h \in \{0, 1, \dots, H\}$ define $T_h = T - h$, $y_{t,h} = y_{t+h}$ and $x_{t,h} = x_t$ for $t = 1, \dots, T_h$. We minimize

$$\mathrm{RSS}(\tau) \;=\; \sum_{h=0}^{H} \sum_{t=1}^{T_h} \big(y_{t,h} - \beta(h)\, x_{t,h}\big)^2$$

over partitions $\tau = (\tau_1, \dots, \tau_{K+1})$ with $0 = \tau_1 < \cdots < \tau_{K+1} = H$ and a discrete-C^1 piecewise quadratic $\beta(\cdot)$.

Segments are $\mathcal{H}_1 = \{0, \dots, \tau_2\}$ and, for $k \geq 2$, $\mathcal{H}_k = \{\tau_k + 1, \dots, \tau_{k+1}\}$. On $\mathcal{H}_k$,

$$\beta(h) = a_k + b_k s + c_k s^2, \qquad s = h - \tau_k.$$

At each interior breakpoint τ_k we impose *value* and *forward-difference* continuity:

$$\beta(\tau_k^-) = \beta(\tau_k^+), \tag{8}$$

$$\Delta\beta(\tau_k) = \Delta\beta(\tau_k + 1), \qquad \Delta\beta(h) = \beta(h) - \beta(h-1). \tag{9}$$

Equivalently, if the preceding segment ends with $\gamma_0 = \beta(\tau_k)$ and $\gamma_1 = \Delta\beta(\tau_k)$, then the next segment satisfies $a_k = \gamma_0$ and $b_k + c_k = \gamma_1$.

Base segment ($k = 1$) For $j \geq 2$, fit $\beta(h) = a_1 + b_1 h + c_1 h^2$ on $h = 0, \dots, j$ by stacked least squares on $x_{t,h}$, $h\, x_{t,h}$, and $h^2 x_{t,h}$. Define

$$C(1, j) = \min_{a_1, b_1, c_1} \sum_{h=0}^{j} \sum_{t=1}^{T_h} \big(y_{t,h} - (a_1 + b_1 h + c_1 h^2) x_{t,h}\big)^2.$$

Store endpoint summaries $\gamma_0(1, j) = a_1 + b_1 j + c_1 j^2$ and $\gamma_1(1, j) = b_1 + c_1(2j - 1)$.

Constrained segments ($k \geq 2$) For $i < j$ with $j - i \geq 3$, let $\Delta = j - i$ and $(\gamma_0, \gamma_1) = (\gamma_0(k-1, i), \gamma_1(k-1, i))$. On $h = i+1, \dots, j$ write $s = h - i$ and fit $\beta(h) = \gamma_0 + \gamma_1 s + c\, s(s-1)$, so only c is free. With $\phi_{t,h} = s(s-1)\, x_{t,h}$ and $r_{t,h} = y_{t,h} - (\gamma_0 + \gamma_1 s) x_{t,h}$,

$$\hat{c} = \frac{\sum \phi_{t,h} r_{t,h}}{\sum \phi_{t,h}^2}, \qquad R = \sum (r_{t,h} - \hat{c}\, \phi_{t,h})^2,$$

where sums are over $t = 1, \dots, T_h$ and $h = i+1, \dots, j$. The endpoint updates are

$$\gamma_0^{\text{new}} = \gamma_0 + \gamma_1 \Delta + \hat{c}\, \Delta(\Delta - 1), \qquad \gamma_1^{\text{new}} = \gamma_1 + 2\hat{c}(\Delta - 1).$$

Bellman recursion and backtracking For $k \geq 2$,

$$C(k, j) = \min_{i \in \{k-1, \dots, j-3\}} \left\{ C(k-1, i) + R(i \to j \mid \gamma_0(k-1, i), \gamma_1(k-1, i)) \right\}.$$

Record the minimizing predecessor $i^\star(k, j)$ and the updated (γ_0, γ_1). Set $\tau_{K+1} = H$ and backtrack $\tau_k = i^\star(k, \tau_{k+1})$ for $k = K, \dots, 2$, with $\tau_1 = 0$. A minimum segment length of $j - i \geq 3$ ensures identification of c after imposing (8)–(9).

The inner transition is computed with two inner products over $\sum_{h=i+1}^{j} T_h$ stacked observations. The number of admissible transitions is

$$O\Big(\sum_{k=2}^{K} \sum_{j=k}^{H} (j - k) \Big) = O(KH^2),$$

which is practical for moderate H and K. In Appendix A we show that, conditional on the breakpoints $\boldsymbol{\tau}$, the segment coefficients solve a linear equality constrained least-squares problem. In particular, the constrained estimator is a linear projection of the unrestricted horizon–by–horizon OLS vector,

$$\widehat{\beta}_{\text{CLS}} = P_K \widehat{\beta}_{\text{OLS}},$$

Algorithm 1: DP for discrete piecewise quadratic impulse responses

Input: Series $y \in \mathbb{R}^T$, $x \in \mathbb{R}^T$; number of segments K; horizon H
Output: Breakpoints $\tau = (\tau_1, \ldots, \tau_{K+1})$ with $\tau_1 = 0$, $\tau_{K+1} = H$

Horizon slices:
for $h = 0$ **to** H **do**
  $T_h \leftarrow T - h; \quad y_{t,h} \leftarrow y_{t+h},$
  $x_{t,h} \leftarrow x_t$ for $t = 1, \ldots, T_h;$

Initialize DP tables:
Set $C[k, j] \leftarrow \infty$ for all $1 \le k \le K, 1 \le j \le H;$
Set $\text{pred}[k, j] \leftarrow \varnothing; \quad g_0[k, j] \leftarrow \text{NaN},$
$g_1[k, j] \leftarrow \text{NaN};$

Base case ($k = 1$):
for $j = 2$ **to** H **do**
  Stack $\{(y_{t,h}, x_{t,h})\}_{t \le T_h,\ h=0,\ldots,j}$ and regress $y_{t,h}$ on $[1,\ h,\ h^2] \cdot x_{t,h};$
  Let $(\hat{a}_1, \hat{b}_1, \hat{c}_1)$ be OLS;
  $C[1, j] \leftarrow \sum_{h=0}^{j} \sum_{t=1}^{T_h} \big(y_{t,h} - (\hat{a}_1 + \hat{b}_1 h + \hat{c}_1 h^2) x_{t,h}\big)^2;$
  $g_0[1, j] \leftarrow \hat{a}_1 + \hat{b}_1 j + \hat{c}_1 j^2; \quad g_1[1, j] \leftarrow \hat{b}_1 + \hat{c}_1(2j - 1);$

Recursive step ($k \ge 2$):
for $k = 2$ **to** K **do**
  for $j = k$ **to** H **do**
    for $i = k - 1$ **to** $j - 3$ **do**
      if $g_0[k-1, i]$ *or* $g_1[k-1, i]$ *is undefined* **then**
        continue
      $\Delta \leftarrow j - i; \quad \gamma_0 \leftarrow g_0[k-1, i]; \quad \gamma_1 \leftarrow g_1[k-1, i];$
      Stack over $h = i + 1, \ldots, j$ with $s \leftarrow h - i$:

$$\phi_{t,h} \leftarrow s(s-1)\, x_{t,h}, \qquad r_{t,h} \leftarrow y_{t,h} - (\gamma_0 + \gamma_1 s)\, x_{t,h}.$$

$$\hat{c} \leftarrow \frac{\sum \phi_{t,h} r_{t,h}}{\sum \phi_{t,h}^2}; \quad R \leftarrow \sum (r_{t,h} - \hat{c}\, \phi_{t,h})^2 \triangleright \texttt{Sums over } t \le T_h,\ h = i+1, \ldots, j$$

      if $C[k-1, i] + R < C[k, j]$ **then**
        $C[k, j] \leftarrow C[k-1, i] + R; \quad \text{pred}[k, j] \leftarrow i;$
        $g_0[k, j] \leftarrow \gamma_0 + \gamma_1 \Delta + \hat{c}\, \Delta(\Delta - 1);$
        $g_1[k, j] \leftarrow \gamma_1 + 2\hat{c}(\Delta - 1);$

Backtracking:
Set $\tau_{K+1} \leftarrow H;$
for $k = K - 1\ 1$ **do**
  $\tau_k \leftarrow \text{pred}[k+1,\ \tau_{k+1}];$
Set $\tau_1 \leftarrow 0$; **return** τ.

where P_K a symmetric idempotent projection matrix that depends on the breakpoint locations. This projection representation is essential for model averaging and inference discussed in the next sections.

3 Jackknife Model Averaging in Local Projections

3.1 Model Averaging

We consider model averaging over a collection of submodels in the local projection (LP) framework. Let $y_t \in \mathbb{R}$ be a scalar outcome and $x_t \in \mathbb{R}$ a scalar regressor (e.g., a shock), observed over $t = 1, \ldots, T$. The impulse response at horizon $h = 0, \ldots, H$ is denoted by $\beta(h)$, and the full vector of impulse responses is $\beta \in \mathbb{R}^{H+1}$. The standard LP model stacks all horizons into the system:

$$Y_t = X_t\beta + \varepsilon_t, \quad t = 1, \ldots, T,$$

where $Y_t = (y_t, y_{t+1}, \ldots, y_{t+H})' \in \mathbb{R}^{H+1}$, $X_t = x_t I_{H+1}$ is a scaled identity matrix, and $\varepsilon_t \in \mathbb{R}^{H+1}$ is a vector of forecast errors. The OLS estimator is:

$$\widehat{\beta}_{\mathrm{OLS}} = \frac{\sum_{t=1}^{T} x_t Y_t}{\sum_{t=1}^{T} x_t^2}.$$

Let $\{P_j\}_{j=1}^{m+1}$ denote a collection of symmetric, idempotent projection matrices on $\mathbb{R}^{H+1}$ representing submodel restrictions (e.g., continuity-constrained piecewise polynomial models), where $P_{m+1} = I_{H+1}$ corresponds to the unrestricted model. Then, for each submodel j, the projected estimator is $\widehat{\beta}_j = P_j\widehat{\beta}_{\mathrm{OLS}}$, and the fitted value at time t is:

$$\widehat{Y}_{j,t} = X_t P_j \widehat{\beta}_{\mathrm{OLS}}.$$

Leave-One-Out Predictions Due to the Kronecker structure of X_t, the leverage of each observation block is constant across horizons and is defined as:

$$h_t = \frac{x_t^2}{\sum_{s=1}^{T} x_s^2}, \quad \text{for each } t.$$

The leave-one-out prediction of Y_t under submodel j is given by:

$$\widehat{Y}_{j,(-t)} = X_t P_j \widehat{\beta}_{\text{OLS},(-t)},$$

where $\widehat{\beta}_{\text{OLS},(-t)}$ is the OLS estimate of β based on the subsample with observation t removed. In Appendix B, we show that the leave-one-out prediction can be expressed as:

$$\widehat{Y}_{j,(-t)} = \frac{1}{1-h_t}\left(\widehat{Y}_{j,t} - h_t P_j Y_t\right),$$

where $\widehat{Y}_{j,t} = X_t P_j \widehat{\beta}_{\text{OLS}}$. This expression provides a more computationally efficient leave-one-out prediction, as it can be computed using only the leverage h_t, without explicitly recomputing the leave-one-out coefficients $\widehat{\beta}_{\text{OLS},(-t)}$. Define the row vector:

$$\psi_{j,t} := \widehat{Y}'_{j,(-t)} \in \mathbb{R}^{1\times(H+1)}.$$

Stacking across submodels yields $\Psi_t \in \mathbb{R}^{(m+1)\times(H+1)}$, whose jth row is $\psi_{j,t}$.

Jackknife Model Averaging Objective The jackknife loss criterion of Hansen and Racine (2012) is:

$$J(w) = \sum_{t=1}^{T} \left\| Y_t - \sum_{j=1}^{m+1} w_j \widehat{Y}_{j,(-t)} \right\|^2 = \sum_{t=1}^{T} \left\| Y_t - w'\Psi_t \right\|^2,$$

where $w \in \mathbb{R}^{m+1}$ is a vector of model weights satisfying $w_j \geq 0$ and $\sum_j w_j = 1$. Expanding the squared error:

$$J(w) = \sum_{t=1}^{T} \left(Y_t' Y_t - 2Y_t' \Psi_t' w + w' \Psi_t \Psi_t' w \right) = w'Aw - 2b'w + \text{const},$$

where:

$$A = \sum_{t=1}^{T} \Psi_t \Psi_t' \in \mathbb{R}^{(m+1)\times(m+1)}, \qquad b = \sum_{t=1}^{T} \Psi_t Y_t \in \mathbb{R}^{m+1}.$$

Explicitly, the matrix entries are:

$$A_{jk} = \sum_{t=1}^{T} \langle \psi_{j,t}, \psi_{k,t} \rangle, \quad b_j = \sum_{t=1}^{T} \langle \psi_{j,t}, Y_t \rangle.$$

Complexity-Penalized Jackknife Averaging We augment the JMA loss function with a complexity penalty proportional to the degrees of freedom of each model:

$$J^p(w) = \sum_{t=1}^{T} \left\| Y_t - \sum_{j=1}^{m+1} w_j \widehat{Y}_{j,(-t)} \right\|^2 + \gamma_T \sum_{j=1}^{m+1} w_j \, \mathrm{df}_j,$$

where $\mathrm{df}_j = \mathrm{tr}(P_j)$ and $\gamma_T = \log(T \cdot (H+1))$. As shown by Zhang and Liu (2019), this additional penalty yields JMA estimators that asymptotically assign weight only to the smallest valid model. This leads to the quadratic program:

$$\min_{w \in \mathbb{R}^{m+1}} \quad w' A w - 2b' w + \gamma \cdot \mathrm{df}' \, w \quad \text{subject to} \quad w \geq 0, \quad \sum_{j=1}^{m+1} w_j = 1. \tag{10}$$

The final JMA estimator is a weighted average of projected LP coefficients:

$$\widehat{\beta}_{\mathrm{JMA}} = \underbrace{\left(\sum_{j=1}^{m+1} w_j^* P_j \right)}_{P(w^*)} \widehat{\beta}_{\mathrm{OLS}},$$

where the weights $\boldsymbol{w}^*$ are the solution to the quadratic program (10). This estimator respects the projection structure imposed by each submodel and leverages leave-one-out predictions to balance bias and variance in finite samples. The model-averaged projection matrix $P(w)$ is a weighted average of the submodel projection matrices P_j, hence the name "averaged projected local projections" (APLP).

3.2 Inference After Model Averaging

We assume that the unrestricted model is a consistent estimator for our impulse response functions:

$$\text{A0. } \hat{\beta} \xrightarrow{p} \beta.$$

Order the restriction projections from more to less restrictive: $P_1, \ldots, P_j, \ldots, P_M$, where M is the number of models and $P_M = I$ gives us back the unrestricted estimator: $P_M\hat{\beta} = \hat{\beta}$. To map this in the framework of Zhang and Liu (2019), we hypothesize that there *might* exist a rank-reducing projection P_{m^*} such that $P_{m^*}\beta = \beta$, which is without loss of generality, because we know it is true for $m^* = M$. Although, in that case, there is no rank reduction. All models estimated under restriction matrices P_j, $j < m^*$ are *too small*, i.e., *underfitted*. Any model with restriction matrix P_k, $k > m^*$ is *overfitted*, but still consistent. We refer to any restriction matrix P_j, $j \geq m^*$ as valid.

We have the following regularity conditions:

C1. $Q_T = \frac{1}{T}X'X \xrightarrow{p} Q = E(X_t'X_t)$, psd.

C2. $Z_t = \frac{1}{\sqrt{T}}X_t'e_t \xrightarrow{d} Z \sim N(0, \Omega)$, where $\Omega = E(X_t'e_te_t'X_t)$

C3. $\bar{h}_T = \max_{1\leq m\leq M} \max_{1\leq i\leq T} h_{ii}^m = o_p(T^{-1/2})$

C4. $\Omega_T = \frac{1}{T}\sum_{t=1}^T X_t'e_te_t'X_t \xrightarrow{p} \Omega$, psd.

From our regularity conditions we get the standard limiting distribution

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} Q^{-1}Z, \qquad Z \sim N(0, \Omega)$$

where $\hat{\beta}$ is the unrestricted OLS estimator.

A direct application of Zhang and Liu (2019)'s Theorem 1 and Theorem 3 under regularity conditions C1, C2 and C3 is that for any $m < m^*$:

$$\hat{w}_{JMA,m} = o_p(T^{-1/2}). \tag{11}$$

Recall we can write any of the restricted LP estimators as $\hat{\beta}^{(j)} = P_j\hat{\beta}$. For any set of *valid* restrictions, we have: $P_j\beta = \beta$. Under that assumption, we can write:

$$\sqrt{T}(\hat{\beta}^{(j)} - \beta) \xrightarrow{d} P_jQ^{-1}Z.$$

Combining this with (11) implies that some weighted average estimator $\hat{\beta}(\hat{w})$ for $\hat{w}_{JMA}$ has the following asymptotic behavior:

$$\begin{aligned}\sqrt{T}(\hat{\beta}(\hat{w}) - \beta) &= \sum_{j=1}^{m^*-1} \hat{w}_j \sqrt{T}(\hat{\beta}_j - \beta) + \sum_{j=m^*}^{M} \hat{w}_j \sqrt{T}(\hat{\beta}_j - \beta) \\ &\to \sum_{s=0}^{S} \tilde{\lambda}_s V_s Z\end{aligned}$$

where λ is a redefined set of weights on the unit simplex that excludes the underfitted models, and $V_s = P_{m^*+s} Q^{-1}$. In particular, for JMA:

$$\tilde{\lambda} = (\tilde{\lambda}_1, ..., \tilde{\lambda}_S) = \text{argmin } \lambda' \Sigma \lambda$$

with Σ an $S \times S$ matrix with $(s, j)^{\text{th}}$ element

$$\Sigma_{sj} = \text{tr}(W_s) + tr(W_j) - Z' V_{\max(s,j)} Z,$$

with $W_s = V_s \Omega$. It follows that the least squares averaging estimator has a non-standard asymptotic distribution when using data-dependent weights, and the distributions are non-pivotal.

We adapt the algorithm in Zhang and Liu (2019) to obtain simulation-based confidence intervals for β as follows.

Step 0. Estimate the full model, estimate $\hat{\sigma}$, $\hat{Q}$ and $\hat{\Omega}$.

Step 1a. Generate R normal random vectors $Z^{(r)}$ of size $H + 1 \times 1$, for $r = 1, ..., R$ with $Z^r \sim N(0, \hat{\Omega})$.

Step 1b. For $m^* = 1, ..., M$, calculate:

- $\hat{V}_s = P_{m^*+s} \hat{Q}^{-1}$ for a given m^*, for $s = 0, ..., S$ with $S = M - m^* + 1$.
- $\hat{\Sigma}^{(r)}(m^*)$ with elements (s, j) given by

$$\hat{\Sigma}^{(r)}_{sj}(m^*) = \text{tr}(W_s) + \text{tr}(W_j) - Z^{(r)\prime} \hat{V}_{\max(s,j)} Z^{(r)},$$

  where we define $W_s = \hat{V}_s \hat{\Omega}$.

- Compute, for λ restricted to the unit simplex,

$$\tilde{\lambda}^{(r)}_{\text{JMA}}(m^*) = \underset{\lambda}{\operatorname{argmin}}\ \lambda' \hat{\Sigma}^{(r)}(m^*)\lambda.$$

Step 1c. Let $\Lambda^{(r)}_{\text{JMA},j}(m^*)$ be the jth component of $\sum_{s=0}^{S} \tilde{\lambda}^{(r)}_{\text{JMA},s}(m^*)\hat{V}_s Z^{(r)}$. We can now compute:

$$\tilde{\Lambda}^{(r)}_{\text{JMA},j}(\tilde{w}_{\text{JMA}}) = \sum_{m^*=1}^{M} \tilde{w}_{\text{JMA},m^*} \Lambda^{(r)}_{\text{JMA},j}(m^*)$$

where $\tilde{w}_{\text{JMA}}$ are the adjusted JMA weights previously defined.

Step 2. Let $\hat{\beta}_j(\hat{w})$ be the jth component of $\hat{\beta}(\hat{w})$, with JMA weights. The confidence interval of β_j is constructed as

$$\left[\hat{\beta}_j(\hat{w}) - T^{-1/2}\hat{q}_j(1-\alpha_2), \quad \hat{\beta}_j(\hat{w}) - T^{-1/2}\hat{q}_j(\alpha/2)\right],$$

where $\hat{q}_j(\alpha/2)$ and $\hat{q}_j(1-\alpha/2)$ are the $(\alpha/2)^{\text{th}}$ and $(1-\alpha/2)^{\text{th}}$ quantiles of $\Lambda^{(r)}_{\text{JMA},j}(\tilde{w}_{\text{JMA}})$.

3.3 Alternative asymptotic framework: local misspecification

An alternative asymptotic framework builds on Liu (2015). In contrast to our baseline setting, where we assume there exists a model size $m^* \leq M$ that coincides with the true model, this framework assumes that there is a model size $m^* < M$ such that all larger models, i.e., all $j \geq m^*$, are *locally misspecified* relative to the true unrestricted model M. In this setting, a consistent estimator of the variance–covariance matrix is obtained from the unrestricted estimator, which can then be used to construct confidence intervals.

While the interval length is unchanged relative to standard local projections, the model-averaged point estimator under this framework achieves a lower mean squared error. Given that local projection estimators tend to undercover in small samples (see, e.g., Herbst and Johannsen 2024), this asymptotic framework can yield more conservative inference with coverage closer to the nominal level, which is of practical interest. The undercoverage problem in local projections is especially pronounced for highly persistent processes, such as AR(1) models with high persistence. The simulation section below demonstrates that this framework can deliver coverage closer to the nominal level in such high-persistence environments.

4 Simulations

This section compares the performance of our proposed estimator with those of existing estimators. The first subsection focuses on three specific DGPs with impulse response functions that have typical shapes often encountered in macroeconomic models and applications. This allows us to illustrate how the method behaves and what its properties are in a controlled environment. In the second subsection, we use the simulation set-up of Li et al. (2024), allowing us to compare the performance of different local projection estimators under hundreds of different data generating processes that mimic the behavior of U.S. macroeconomic and financial data.

Existing methods The simulations compare our APLP estimator to the standard local projection estimator (estimated using least squares, hence OLS), and two alternative local projection estimators that have been proposed in the literature. Firstly, bias-corrected local projection (BCLP) as in Herbst and Johannsen (2024), who, using a Nagar (1959) expansion, derive an explicit expression for small sample bias in local projections. Since the small sample bias shrinks to zero at a linear rate, the asymptotic variance-covariance matrix of the estimator coincides with that of OLS. Second, smooth local projections (SLP) as in Barnichon and Brownlees (2019), who penalize deviations of the LP estimator from B-splines. For inference, they propose a *heuristic* variance-covariance estimator, because cross-validation of the shrinkage parameter introduces a non-standard testing problem (e.g., studied in McCloskey, 2017). For details on the methods, we refer to their papers.

4.1 Three Data Generating Processes

Simulation set-up In this subsection, we show simulation results for three different ARMA(p,q) processes with impulse response functions that have shapes often encountered in macroeconomic models and applications. We consider simulation lengths of both $T = 100$ and $T = 200$.

DGP 1: AR(1): $y_t = \rho y_{t-1} + \varepsilon_t$, with $y_t^{\text{obs}} = y_t + \eta_t$. We parametrize: $\rho = 0.8$, $\varepsilon_t \sim N(0,1)$ and $\eta_t \sim N(0,s^2)$ with s^2 equal to one times the unconditional variance of y_t.

DGP 2: ARMA(1,12): $y_t = \rho y_{t-1} + \varepsilon_t + \sum_{l=1}^{12} \theta_l \varepsilon_{t-l}$, with $y_t^{\text{obs}} = y_t + \eta_t$. We parametrize this process to give rise to a steep positive hump-shaped impulse response function followed by a gradual negative hump: $\rho = 0.8$, $\varepsilon_t \sim N(0,1)$, $\eta_t \sim N(0, s^2)$, with s^2 one times the unconditional variance of y_t and $\theta = [1,\ \frac{1}{2},\ \frac{1}{4},\ 0,\ -\frac{1}{4},\ -\frac{1}{2},\ -\frac{6}{14},\ -\frac{5}{14},\ -\frac{4}{14},\ -\frac{3}{14},\ -\frac{2}{14},\ -\frac{1}{14}]$.

DGP 3: ARMA(3,5): $y_t = \sum_{j=1}^{3} \rho_j y_{t-j} + \varepsilon_t + \sum_{l=1}^{5} \theta_l \varepsilon_{t-l}$, with $y_t^{\text{obs}} = y_t + \eta_t$ We parametrize this process to generate an impulse response function with positive and persistent hump-shaped response, by choosing: $\rho = [1.2,\ -0.6,\ 0.2]$, $\theta = [0.5,\ 0.5,\ 0.5,\ 0.25,\ 0.1]$. We have $\varepsilon_t \sim \mathcal{N}(0,1)$, $\eta_t \sim \mathcal{N}(0, s^2)$ with s^2 equal to two times the unconditional variance of y_t.

The implied impulse response functions of these three DGPs are shown in Figures 1a-1c, respectively, in the first panel with the dotted black line.

We simulate 1,000 datasets from the DGPs described above, for both $T = 100$, $T = 200$. Since we are also interested in the asymptotic behavior of the model weights, we present those for $T = 500$. $T = 100$ is close to the median sample size used in local projections according to Herbst and Johannsen (2024), and allows us to study small-sample behavior of the estimators. $T = 200$ is still well within the range of local projection set-ups often used by researchers.

Metrics Denote the true IRF as θ_h^0 for $h = 0, \ldots, H_{\max}$. We consider:

- The average point estimate at each horizon: $\bar{\hat{\theta}}_h = \frac{1}{N_{MC}} \sum_{i=1}^{N_{MC}} \hat{\theta}_h^i$ for $h = 0, \ldots, H_{\max}$,
- The variance of the estimator at each horizon: $\frac{1}{N_{MC}} \sum_{i=1}^{N_{MC}} (\hat{\theta}_h^i - \bar{\hat{\theta}}_h)^2$ for $h = 0, \ldots, H_{\max}$,
- The absolute bias at each horizon: $|\bar{\hat{\theta}}_h - \theta_h^0|$,
- The Mean Squared Error (MSE) of the estimator at each horizon: $\frac{1}{N_{MC}} \sum_{i=1}^{N_{MC}} (\hat{\theta}_h^i - \theta_h^0)^2$ for $h = 0, \ldots, H_{\max}$,
- Pointwise confidence interval length (95%) relative to the OLS LP estimator
- Pointwise coverage probabilities of the 95% confidence intervals

In addition to looking at pointwise inference, since an impulse response function is a vector-valued outcome, we also look at simultaneous confidence intervals, as in

Montiel Olea and Plagborg-Møller (2019). We compare sup-t and Bonferroni bands for standard OLS LP's and bias-corrected LP's with Bonferroni bands for SLP and APLP. We look at coverage, which in this setting is defined as the probability that the entire impulse response falls into the confidence set, and average length.

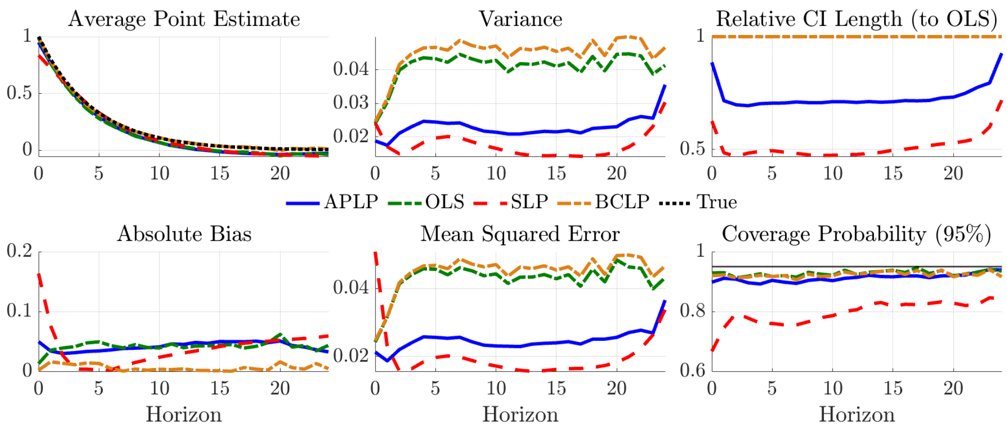
Results Figure 1 summarizes the simulation results for the key metrics discussed above, for $T = 100$. Figure A1 in the Appendix reports the same metrics for $T = 200$. The following results are worth highlighting. SLP can be severely biased, especially at short horizons. APLP is less biased than SLP, and for moderate to long horizons, the bias is comparable to that of OLS. Note that OLS is not unbiased because of small sample bias, as documented in Herbst and Johannsen (2024). The bias-corrected LP estimator from Herbst and Johannsen (2024) has overall lowest bias, but OLS dominates BCLP in terms of MSE. APLP typically has the lowest MSE for shorter horizons, while SLP has the lowest MSE at longer horizons.

When evaluating an estimation method, one does not only care about point estimation, but also inference. The third column in Figure 1 shows that across all three DGPs, APLP has coverage comparable to OLS and BCLP. It does so with confidence intervals that are about 20% less wide. The SLP confidence intervals undercover more than the other methods, which is a consequence of both SLP ignoring the uncertainty in the shrinkage parameter selection during cross-validation, and the large bias it introduces at short horizons. Hence, while SLP confidence bands are shorter, they seem unreliable for inference.

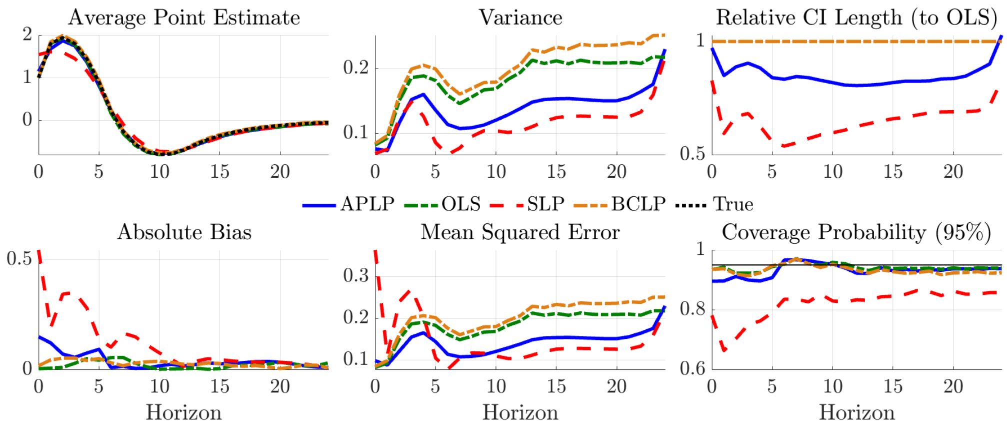
Next, we turn to simultaneous inference. Tables 1-3 summarize the coverage and average length of the Bonferroni and sup-t confidence bands for OLS and BCLP, and the Bonferroni confidence bands for SLP and APLP. The length is computed as the average length of the confidence bands across horizons, relative to the OLS Bonferroni bands. Three results stand out. Firstly, the sup-t confidence bands, while indeed slightly narrower than the Bonferroni bands, tend to undercover, especially for $T = 100$, suggesting that in finite-samples, it may be more prudent to use Bonferroni bands instead. Second, coverage of SLP is poor, consistent with the results in Figure 1. Third, for these three DGPs, APLP has better simultaneous coverage properties than OLS and BCLP, while drastically reducing the length of the set, with average reductions between 12-24%.

Figure 1: Comparison of methods for DGPs with T = 100

(a) DGP 1: AR(1)



(b) DGP 2: ARMA(1, 12)



(c) DGP 3: ARMA(3, 5)

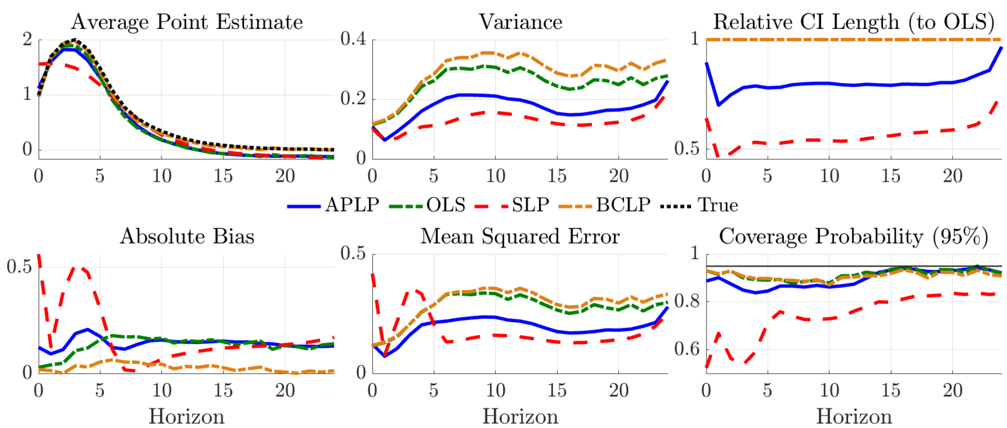


Table 1: Coverage and length of simultaneous confidence bands for DGP 1

Method	Level	Coverage				Average rel. length			
		Bonf.	sup-t	Bonf.	sup-t	Bonf.	sup-t	Bonf.	sup-t
		$T = 100$		$T = 200$		$T = 100$		$T = 200$	
OLS	90 %	77.4	75.2	87.8	85.2	1.00	0.98	1.00	0.98
	95 %	86.6	85.6	93.2	92.3	1.00	0.99	1.00	0.99
BCLP	90 %	75.8	73.6	87.3	84.7	1.00	0.98	1.00	0.98
	95 %	84.1	82.6	92.6	91.8	1.00	0.99	1.00	0.99
SLP	90 %	46.0	–	69.6	–	0.51	–	0.62	–
	95 %	54.0	–	77.3	–	0.51	–	0.62	–
APLP	90 %	87.0	–	90.3	–	0.76	–	0.76	–
	95 %	91.0	–	93.7	–	0.76	–	0.77	–

Table 2: Coverage and length of simultaneous confidence bands for DGP 2

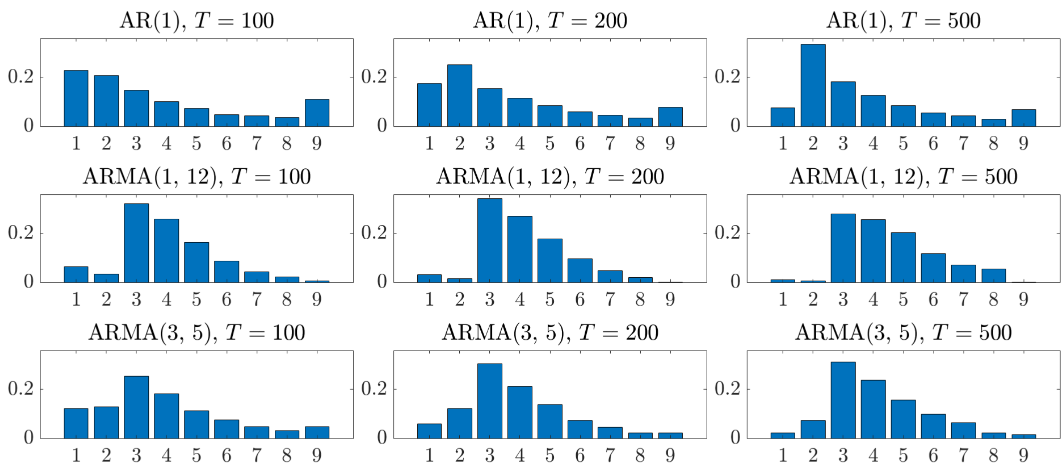
Method	Level	Coverage				Average rel. length			
		Bonf.	sup-t	Bonf.	sup-t	Bonf.	sup-t	Bonf.	sup-t
		$T = 100$		$T = 200$		$T = 100$		$T = 200$	
OLS	90 %	87.7	83.5	91.0	88.9	1.00	0.95	1.00	0.96
	95 %	92.8	90.8	94.4	93.1	1.00	0.97	1.00	0.97
BCLP	90 %	84.9	80.8	89.6	86.6	1.00	0.95	1.00	0.96
	95 %	90.5	88.0	93.9	91.9	1.00	0.97	1.00	0.97
SLP	90 %	49.0	–	64.0	–	0.65	–	0.76	–
	95 %	56.7	–	70.4	–	0.65	–	0.76	–
APLP	90 %	88.7	–	91.3	–	0.87	–	0.88	–
	95 %	92.0	–	95.3	–	0.88	–	0.88	–

Table 3: Coverage and length of simultaneous confidence bands for DGP 3

Method	Level	Coverage (%)				Average rel. length			
		Bonf.	sup-t	Bonf.	sup-t	Bonf.	sup-t	Bonf.	sup-t
		$T = 100$		$T = 200$		$T = 100$		$T = 200$	
OLS	90 %	76.4	73.0	88.3	85.0	1.00	0.97	1.00	0.97
	95 %	84.4	81.9	92.1	90.9	1.00	0.98	1.00	0.98
BCLP	90 %	75.6	71.4	86.3	84.1	1.00	0.97	1.00	0.97
	95 %	83.4	80.8	91.1	90.0	1.00	0.98	1.00	0.98
SLP	90 %	24.4	–	44.5	–	0.56	–	0.68	–
	95 %	30.2	–	52.6	–	0.56	–	0.68	–
APLP	90 %	80.1	–	89.2	–	0.82	–	0.83	–
	95 %	84.5	–	93.1	–	0.82	–	0.84	–

At last, Figure 2 shows the average model weights across simulations that the JMA criterion places on the different model sizes. For $H_{\max} = 24$, the largest (hence fully unrestricted) model has index 9. For DGP 1, the method places weight on both small and large models. Interestingly, for DGP 2 and DGP 3, we see that the method places no weight on both the models that are too large and too small, consistent with our asymptotic theory, as well as with our motivation that for some impulse response functions, a lower rank representation may exist.

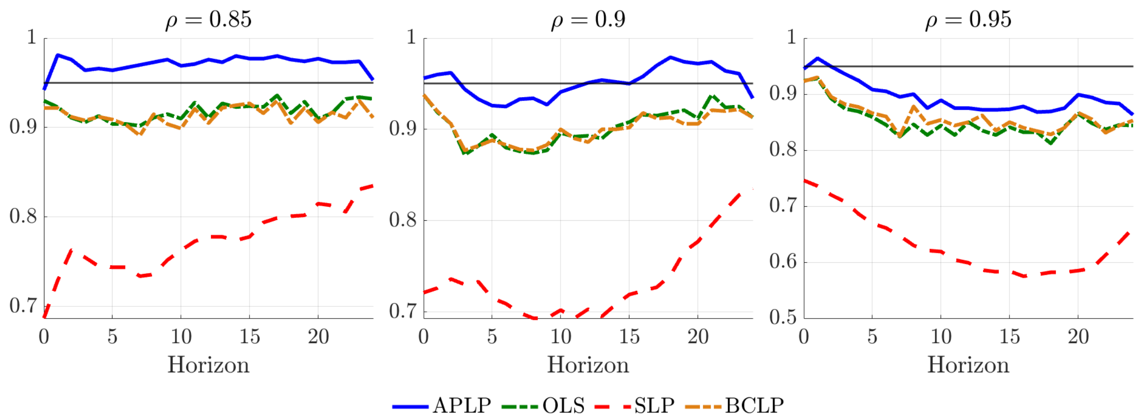
Figure 2: Average model weights



4.2 Coverage under alternative asymptotic framework

In this subsection, we show a subset of simulation results under the alternative asymptotic framework discussed in Section 3.3. We focus on AR(1) processes with three different levels of persistence, for $T = 100$. The focus on AR(1) is motivated by the weights in Figure 2 that suggest that even at $T = 500$, for persistent AR(1) processes, JMA places considerable weight on the largest model. Hence, in Figure 3, we compare coverage when using the asymptotic variance-covariance matrix of the unrestricted model instead to construct confidence intervals. The figure shows that this alternative asymptotic framework allows for improved coverage in highly persistent environment with small samples, which is a particular setting where standard local projections undercover. Importantly, it can do so with confidence intervals as wide as standard local projections, but by centering the confidence interval around the APLP estimate instead.

Figure 3: Coverage (95%) under the alternative asymptotic framework consistent with Liu (2015) for an AR(1) with varying persistence and $T = 100$



4.3 Hundreds of Data Generating Processes

Next, we benchmark the performance of our method on hundreds of random data generating processes. More specifically, we follow Li et al. (2024) who build thousands of random data-generating processes from a non-stationary dynamic factor model (DFM) calibrated to the large-scale U.S. macroeconomic dataset of Stock and Watson (2016).

Data Generating Process The DFM set-up assumes that a large-dimensional vector of observed macroeconomic variables X_t is driven by a low-dimensional vector of latent factors f_t and idiosyncratic components ν_t. The latent factors follow a non-stationary VECM with a VAR(p_f) representation:

$$f_t = \Phi(L) f_{t-1} + H\varepsilon_t, \quad \varepsilon_t \sim \text{i.i.d.}(0, I_{n_f}), \tag{12}$$

where f_t is $n_f \times 1$, ε_t is the $n_f \times 1$ vector of aggregate shocks, and H is $n_f \times n_f$ and determines the impact impulse responses of the factors to the shocks.

The observed macroeconomic series follow as:

$$X_t = \Lambda f_t + \nu_t, \tag{13}$$

where ν_t is the $n_X \times 1$ idiosyncratic component, satisfying a potentially non-stationary AR(p_ν) process:

$$\nu_{i,t} = \Gamma_i(L)\nu_{i,t-1} + \Xi_i \xi_{i,t}, \quad \xi_{i,t} \text{ i.i.d. across } t, i. \tag{14}$$

All shocks and innovations are assumed to be jointly normal and homoskedastic. Li et al. (2024) estimate all parameters of this model, except for impact matrix H, which follows from the procedure discussed below.

We focus on the set-up where the econometrician observes, without measurement error:

$$w_t = (\varepsilon_{1,t}, \bar{w}_t)',$$

where $\varepsilon_{1,t}$ is the structural shock and $\bar{w}_t$ is a subset of observed macroeconomic variables drawn from X_t. In this case, the object of interest is the impulse response of an outcome variable $y_t \in \bar{w}_t$ to a one-standard-deviation innovation in $\varepsilon_{1,t}$, given by:

$$\theta_h = \bar{\Lambda}_{l_y,\cdot} \Theta^f_{\cdot,1,h}, \quad h = 0, 1, 2, \ldots, \tag{15}$$

where $\Theta^f(L)$ are the factor responses to shocks, $\bar{\Lambda}$ selects rows corresponding to $\bar{w}_t$, and l_y indexes the position of y_t in $\bar{w}_t$.

Simulation set-up As in Li et al. (2024), we draw many lower-dimensional DGPs from the large-scale DFM using the following procedure. Each DGP contains $n_{\bar{w}} = 5$ macroeconomic variables. Li et al. (2024) distinguishes between two types of DGPs: monetary policy (MP) and government spending (G) DGPs. The former always includes the federal funds rate in $\bar{w}_t$ as policy variable, the latter always includes federal government spending. The structural impact response matrix H is selected so as to maximize the impact effect of the shock $\varepsilon_{1,t}$ on the federal funds rate (for monetary shocks) and government spending (for fiscal shocks), subject to the constraint that H is consistent with Li et al. (2024)'s estimate of the reduced-form innovation variance–covariance matrix for the factors. The remaining four variables in $\bar{w}_t$ are chosen randomly from X_t, in such a way that it includes always at least one variable measuring real activity, and at least one variable measuring prices. The relevant impulse response variable y_t is chosen randomly from the four non-policy variables. The results below are based on 70 MP DGPs and 70 G DGPs.

Results Figure 4 summarizes the simulation results. Panel 4a reports the median absolute bias of the estimated IRF at each horizon (across DGPs). Consistent with Li et al. (2024), bias-corrected LPs (Herbst and Johannsen, 2024) reduce bias relative to standard LPs. This gain comes at the cost of higher variance (Panel 4b), which lowers coverage relative to LPs (Panel 4c) because the BCLP confidence intervals simply re-center those from the standard LP without adjusting their width.

Smooth local projections (Barnichon and Brownlees, 2019) introduce substantial bias at short horizons (Panel 4a) but reduce variance (Panel 4b). However, their median coverage is poor, roughly 0.15 points lower than the other methods at horizon 7 for example, due to both narrow confidence intervals that ignore uncertainty from cross-validation and large bias.

APLP does not introduce much bias compared to standard LPs. The median absolute bias is a touch larger at shorter horizons but typically smaller beyond horizon 10. More importantly, APLP achieves about a 20% reduction in variance for horizons 3 to 19, as shown in Panel 4b. The resulting median confidence intervals tend to be about 20% narrower than those of both LP and BCLP (Panel 4d), while maintaining coverage similar to standard LPs (Panel 4c).

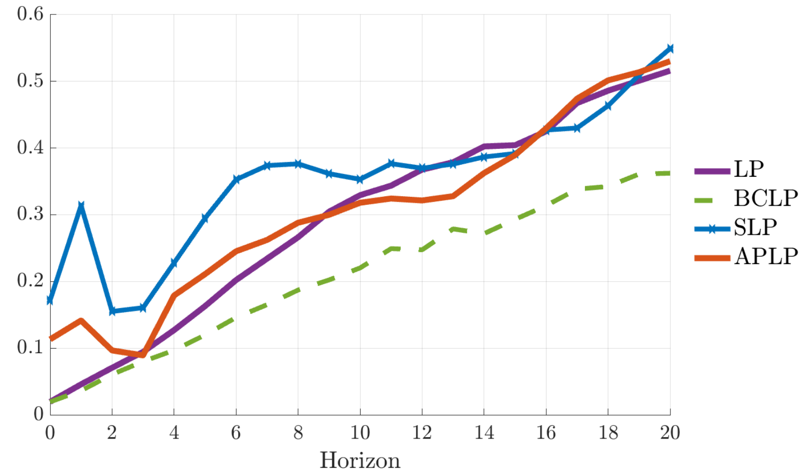
Lastly, Table 4 shows the median (across DGPs) coverage of Bonferroni simultaneous confidence intervals. APLP achieves better median coverage than the other methods. SLP simultaneous confidence bands severely undercover. These results show Tables 1-3 generalize.

Table 4: Median (across DGPs) coverage of Bonferroni simultaneous confidence intervals

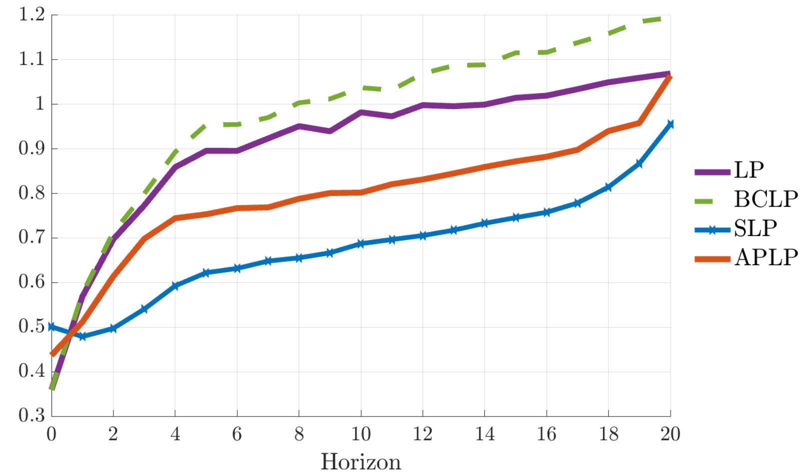
DGP	Level	LP	BCLP	SLP	APLP
MP	90%	71.3	69.8	43.4	73.8
	95%	77.9	77.0	50.4	79.1
G	90%	77.6	74.3	54.2	80.8
	95%	84.5	81.4	61.5	86.8
Both	90%	74.1	71.4	48.2	78.8
	95%	81.9	79.0	55.1	85.1

Figure 4: Li et al. (2024) simulation framework: comparing methods across hundreds of DGPs

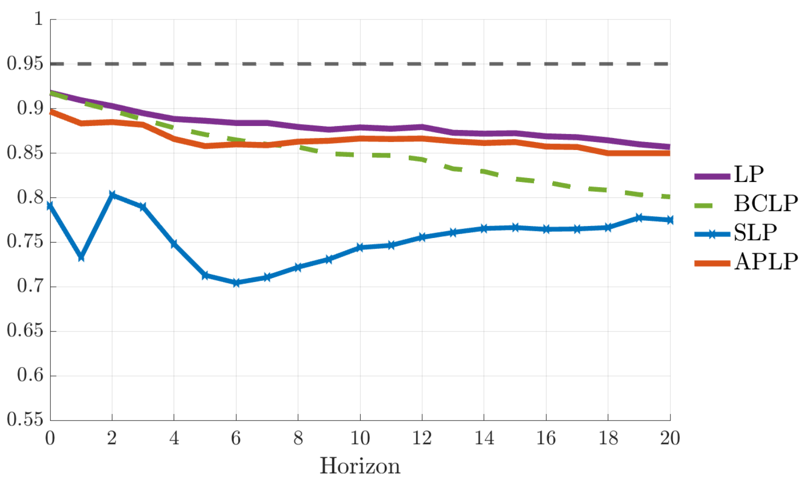
(a) Median (across DGPs) of absolute bias $|\mathbb{E}(\hat{\theta}_h - \theta_h)|$ of the different estimation procedures, relative to $\sqrt{\frac{1}{21}\sum_{h=0}^{20}\theta_h^2}$



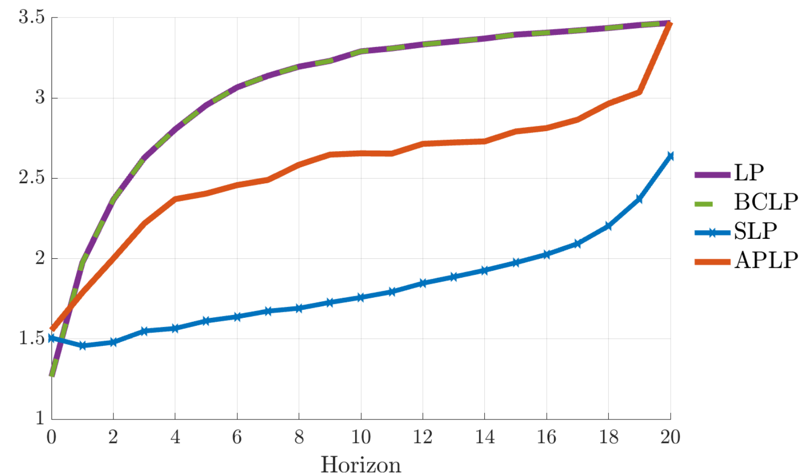
(b) Median (across DGPs) of standard deviation $\sqrt{\mathrm{Var}(\hat{\theta}_h)}$ of the different estimation procedures, relative to $\sqrt{\frac{1}{21}\sum_{h=0}^{20}\theta_h^2}$



(c) Median (across DGPs) coverage (relative to 95% CI)



(d) Median (across DGPs) of 95% CI Width, relative to $\sqrt{\frac{1}{21}\sum_{h=0}^{20}\theta_h^2}$



5 Applications: Financial Markets and Monetary Policy

5.1 Application 1: Risk Premia and Monetary Policy

In this first application, we revisit the empirical evidence in Kekre and Lenel (2022), who document how risk premia move in response to monetary policy shocks. Kekre and Lenel (2022) use an SVAR, we use local projections. We compare the impulse response function estimates obtained using least squares, our proposed average projection method (APLP), and smooth local projections. The main objective of the application is to show how point estimates and confidence intervals vary across methods and different outcome variables.

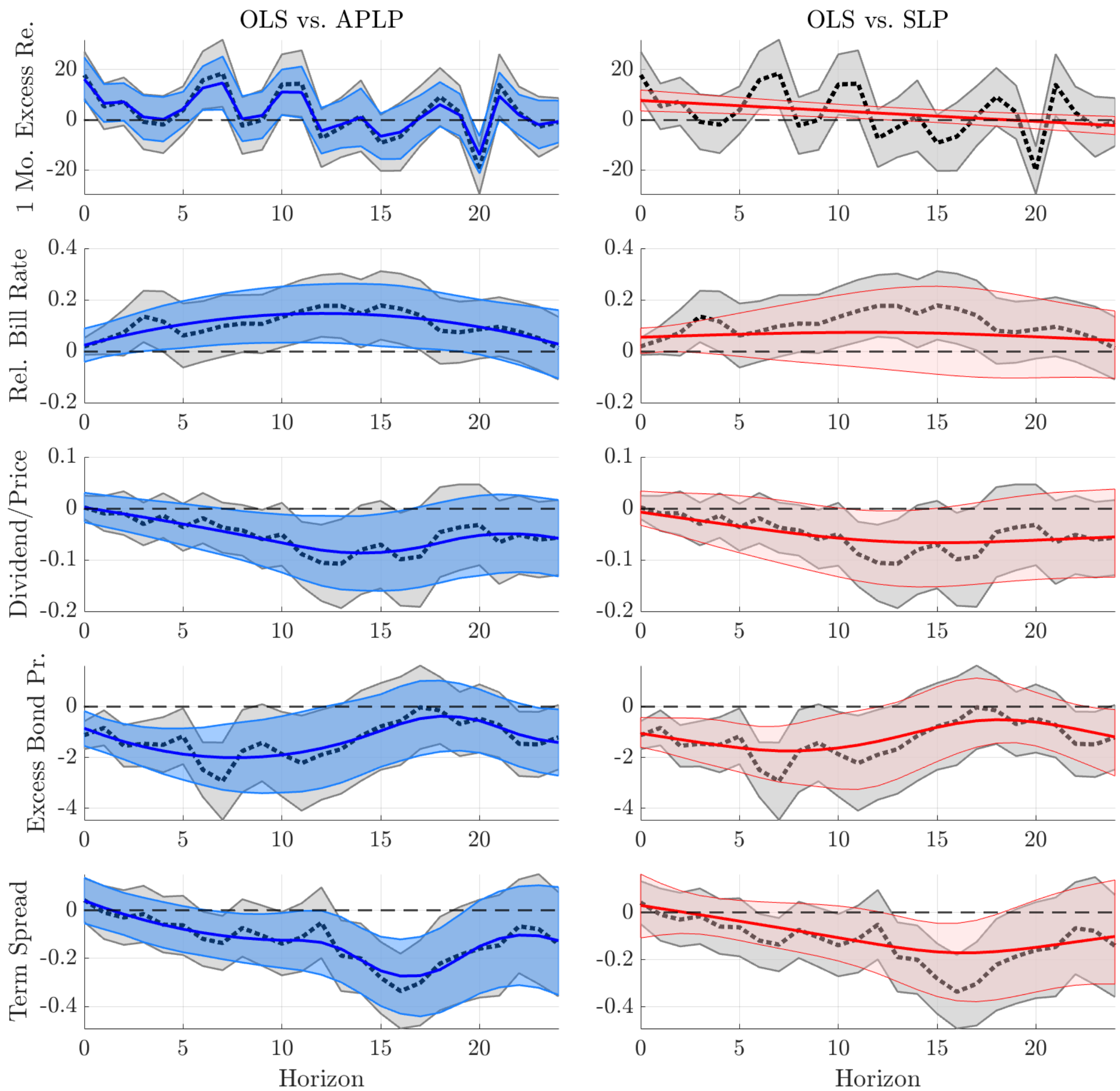
Set-up and data We use the high-frequency identification monetary policy shock series of Gertler and Karadi (2015) as our proxy. We look at the following outcome variables: (i) the one-month excess return, defined as the S&P 500 return relative to the 1-month T-bill, (ii) the relative bill rate, defined as the three-month Treasury rate minus its twelve-month moving average, (iii) the dividend price ratio, smoothed as in Kekre and Lenel (2022),[1] (iv) the excess bond premium from Gilchrist and Zakrajšek (2012) and (v) the term spread, which is the difference between the three month and ten year Treasury yield, also often used as a proxy for the slope of the yield curve.

As controls, we include current and lagged values of the log growth rate of industrial production, the log growth rate of CPI inflation, and the excess bond premium. In addition, we control for own-lags and lags of the shock. For all specifications shown, we use twelve lags, corresponding to a full year. The sample covers January 1991 through June 2012 and the data has a monthly frequency. All data is obtained from the replication package of Kekre and Lenel (2022).

Results The resulting impulse responses are shown in Figure 5 with 90% confidence intervals. First, focusing on OLS and APLP in the first column, we find that the OLS and APLP estimates tend to closely agree, and the results are broadly

[1] In Kekre and Lenel (2022), the dividend-price ratio is computed as the three-month moving average of dividends divided by the price of the stock at the end of the month, value-weighted over the S&P 500.

Figure 5: Impulse responses to the Gertler and Karadi (2015) shock, 90% confidence intervals – OLS in grey with dotted line, APLP in blue, SLP in red.



consistent with the SVAR estimates in Kekre and Lenel (2022).[2] As expected, APLP produces smoother point estimates than OLS, but follow the OLS estimates closely. The APLP confidence bands tend to be less wide than the OLS confidence bands. Consequently, at the 90% confidence level, the relative bill rate impulse response is now significantly different from zero from five to eighteen months after the shock, the dividend/price ratio from seven until sixteen months, and the term spread six

[2]Compare to Figure 1 of their paper. For scale, multiply the estimates of Kekre and Lenel (2022) by 10.

until nineteen months after the shock impact, hence improving both inference on the sign and duration of the effect.

The second column of Figure 5 presents the results from the Smooth Local Projections (SLP) method, where the shrinkage parameter is selected via cross-validation, following the procedure outlined in Barnichon and Brownlees (2019). Compared to OLS, SLP frequently yields narrower confidence intervals due to the smoothing and regularization it imposes. However, this comes at the cost of more aggressive shrinkage, which occasionally leads to 90% confidence intervals that encompass zero, while they did not for OLS, consider, for example, the relative bill rate. For the one-month excess return, cross-validation with SLP selects a particularly high shrinkage parameter, resulting in a flat line and 90% confidence intervals that often exclude the OLS point estimates.

5.2 Application 2: Forward Guidance and the Yield Curve

In the second application, we analyze the effects of *standard* Federal Funds Rate (FFR) shocks and *unconventional* Forward Guidance shocks, as identified in Swanson (2021), on the Yield curve. The objective of this application is two-fold. Firstly, Swanson (2021) uses local projections to assess the effects of its shocks on U.S. Treasury yields of different maturities, but only looks at horizons up to approximately four months. We show that while at short horizons, the effects of FFR shocks and Forward Guidance shocks may look similar, Forward Guidance shocks have important effects at longer horizons. Second, we want to show that these effects are present even when we introduce temporal aggregation, that is, when we estimate the local projections at the weekly, monthly or even quarterly frequency.

Set-up and data We use the daily FFR and Forward Guidance shock series of Swanson (2021). In addition, in all regressions, we control for the large-scale asset purchases (LSAP) shock identified by Swanson (2021).[3] To aggregate the shocks up to weekly, monthly and quarterly, we take the sum over that respective period.

Our outcome variables of interest are U.S. Treasury yields at different maturities, in particular, the three-month yield, the one-year yield and the ten-year yield.[4] When aggregating to the weekly, monthly and quarterly frequency, we use the yield at

[3]All obtained from the replication package of Swanson (2021) on August 3, 2025.

[4]Obtained from https://www.federalreserve.gov/releases/h15/ on August 3, 2025.

the end of that respective period. A standard way to represent the yield curve is to focus on: (i) level, (ii) slope and (iii) curvature. Standard empirical proxies for these objects (e.g., Diebold et al., 2006) are: (i) the level is typically measured as the ten-year yield, (ii) the slope as the difference between the ten-year yield and the three-month yield, and (iii) the curvature is defined as the difference between the ten-year yield and the one-year yield, and the difference between the one-year yield and the three-month yield. These are also the definitions we adopt below.

We adopt the same regression specifications as in Swanson (2021):

$$y_{t+h} - y_t = \beta_h x_t + \gamma_h w_t + \varepsilon_t^h,$$

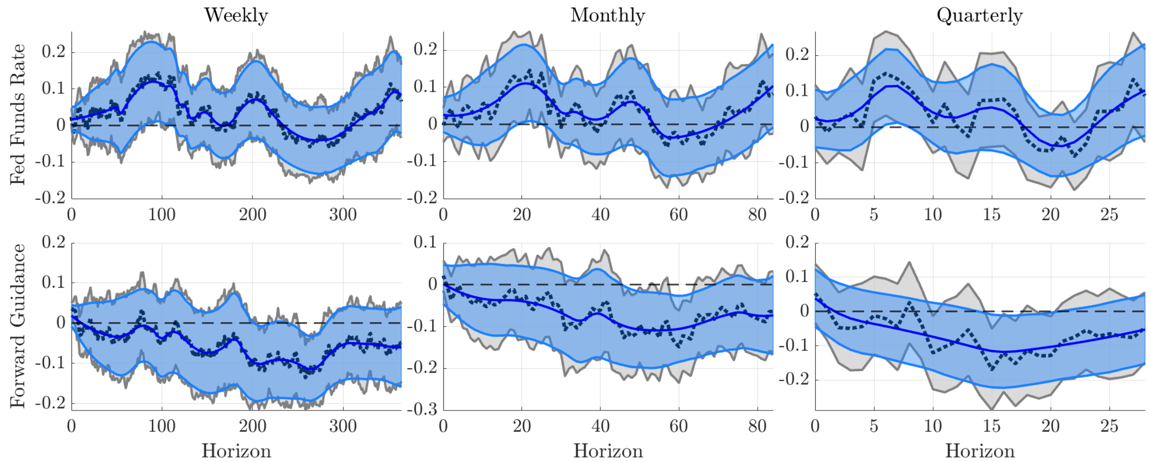
where x_t is either the FFR shock or the Forward Guidance shock, and we control for the other shock not included and the LSAP shock.

Results Figure 6 shows the estimated impulse response functions for OLS and our proposed average projection method. The first panel shows the effects on the level, in the first row the FFR shock, and in the second row the Forward Guidance shock. The FFR shock increases the level, and this effect is significant at 90% after around 1.5 years after the shock for several periods. The Forward Guidance shock lowers the level, and this is significant at the 90% level after ca. 4 years.

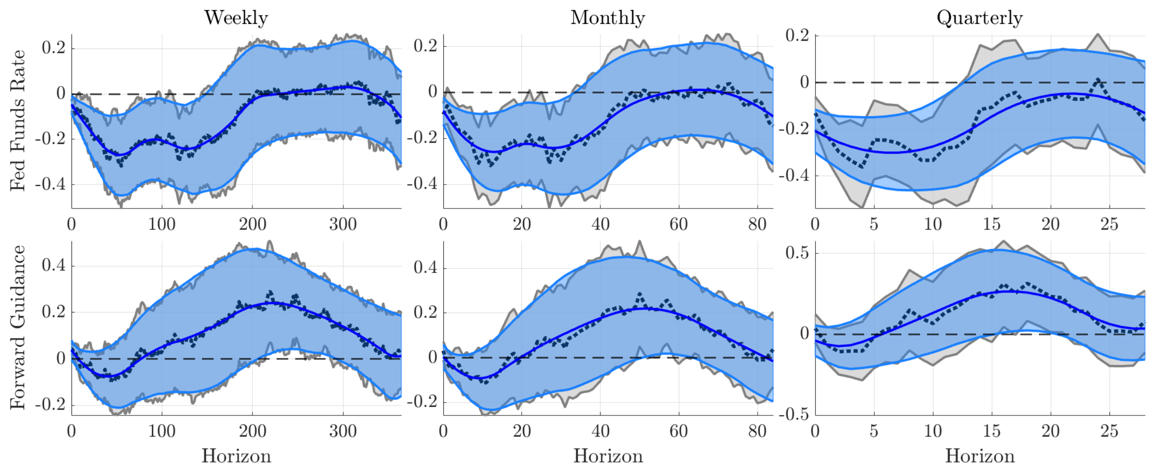
The first row of panel (b) of Figure 6 shows the effects of a FFR shock on the slope of the yield curve, and it lowers the slope significantly for ca. 3 years. Note that these effects are consistent in size and shape with the Gertler and Karadi (2015) shock in Figure 5 (see the effect on Term premium). A Forward Guidance shock, on the other hand, initially lowers the slope, but not significantly, and increases the slope, and this is significant at the 90% level around ca. 5 years after the shock. Similarly but reversed, in panel (c), an FFR shock increases the curvature, and this dies down after ca. four years. The Forward Guidance shock initially increases the curvature, but significantly lowers it five years after impact.

Figure 6: Fed Funds Shocks versus Forward Guidance – Effects on the Yield Curve (90% CI)

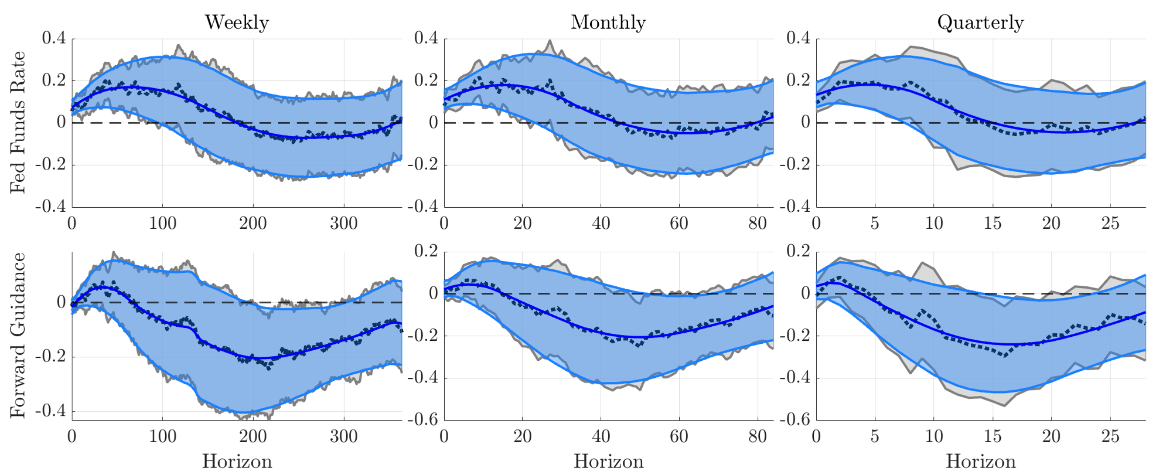
(a) Level



(b) Slope



(c) Curvature



The estimates are broadly consistent across the different levels of temporal aggregation, but, as one would expect, the APLP smooths more when there are fewer observations, and the difference between the OLS and APLP point estimates and confidence intervals is largest at the quarterly frequency, showing the value of APLP when doing inference in smaller samples.

6 Conclusion

We propose a novel nonparametric estimator for functions with discrete ordered support, with impulse response function estimation as the leading example. Our main asymptotic framework assumes that a lower-dimensional representation of the impulse response function may exist in the form of a piecewise continuous function. In this case, the method asymptotically selects this representation, yielding an estimator that is more efficient than the standard local projection approach of Jordà (2005). If no such representation exists, the estimator converges asymptotically to the local projection estimator. Simulation results demonstrate that the method introduces little bias while substantially reducing variance relative to standard local projections. Inference accounts for model uncertainty, producing confidence intervals that are meaningfully shorter while coverage is comparable to that of standard local projections. Two empirical applications illustrate the method's practical value, yielding smoother point estimates and tighter confidence intervals than local projections, particularly in settings with noisy data or smaller samples.

It is well known that local projections can have poor small sample properties, and, in particular, confidence intervals can severely undercover (Herbst and Johannsen, 2024). Hence, as a complementary perspective, we also study an alternative asymptotic framework in which we assume the lower-dimensional models are locally misspecified relative to the larger model. In that case, the variance–covariance matrix coincides with that of the unrestricted model. Also under this framework, the method is valuable: the point estimator lowers mean squared error, and simulations show coverage close to nominal levels, even in highly persistent environments with small samples, where standard local projection confidence intervals undercover. This feature is attractive, as it means we can improve coverage of impulse response function estimates in finite samples, without widening the confidence intervals relative to standard local projections.

References

Bai, J. and P. Perron (2003): "Computation and Analysis of Multiple Structural Change Models," *Journal of Applied Econometrics*, 18, 1–22.

Barnichon, R. and C. Brownlees (2019): "Impulse Response Estimation by Smooth Local Projections," *Review of Economics and Statistics*, 101, 522–530.

Barnichon, R. and C. Matthes (2018): "Functional Approximation of Impulse Responses," *Journal of Monetary Economics*, 99, 41–55.

Diebold, F. X., G. D. Rudebusch, and S. B. Aruoba (2006): "The Macroeconomy and the Yield Curve: a Dynamic Latent Factor Approach," *Journal of econometrics*, 131, 309–338.

Fearnhead, P., R. Maidstone, and A. Letchford (2019): "Detecting Changes in Slope With an L0 Penalty," *Journal of Computational and Graphical Statistics*, 28, 265–275.

Ferreira, L. N., S. Miranda-Agrippino, and G. Ricco (2025): "Bayesian Local Projections," *Review of Economics and Statistics*, 1–15.

Gertler, M. and P. Karadi (2015): "Monetary Policy Surprises, Credit Costs, and Economic Activity," *American Economic Journal: Macroeconomics*, 7, 44–76.

Gilchrist, S. and E. Zakrajšek (2012): "Credit Spreads and Business Cycle Fluctuations," *American economic review*, 102, 1692–1720.

Hansen, B. E. and J. S. Racine (2012): "Jackknife Model Averaging," *Journal of Econometrics*, 167, 38–46.

Herbst, E. P. and B. K. Johannsen (2024): "Bias in Local Projections," *Journal of Econometrics*, 240, 105655.

Ho, P., T. A. Lubik, and C. Matthes (2024): "Averaging Impulse Responses Using Prediction Pools," *Journal of Monetary Economics*, 146, 103571.

Jordà, Ò. (2005): "Estimation and Inference of Impulse Responses by Local Projections," *American Economic Review*, 95, 161–182.

Kekre, R. and M. Lenel (2022): "Monetary Policy, Redistribution, and Risk Premia," *Econometrica*, 90, 2249–2282.

Kleinberg, J. and E. Tardos (2006): *Algorithm design*, Pearson.

Li, D., M. Plagborg-Møller, and C. K. Wolf (2024): "Local Projections vs. VARs: Lessons from Thousands of DGPs," *Journal of Econometrics*, 244, 105722.

Liu, C.-A. (2015): "Distribution Theory of the Least Squares Averaging Estimator," *Journal of Econometrics*, 186, 142–159.

McCloskey, A. (2017): "Bonferroni-based Size-correction for Nonstandard Testing Problems," *Journal of Econometrics*, 200, 17–35.

Montiel Olea, J. L. and M. Plagborg-Møller (2019): "Simultaneous Confidence Bands: Theory, Implementation, and an Application to SVARs," *Journal of Applied Econometrics*, 34, 1–17.

Nagar, A. L. (1959): "The Bias and Moment Matrix of the General K-Class Estimators of the Parameters in Simultaneous Equations," *Econometrica: Journal of the Econometric Society*, 575–595.

Stock, J. H. and M. W. Watson (2016): "Dynamic Factor Models, Factor-Augmented Vector Autoregressions, and Structural Vector Autoregressions in Macroeconomics," in *Handbook of macroeconomics*, Elsevier, vol. 2, 415–525.

Swanson, E. T. (2021): "Measuring the Effects of Federal Reserve Forward Guidance and Asset Purchases on Financial Markets," *Journal of Monetary Economics*, 118, 32–53.

Zhang, X. and C.-A. Liu (2019): "Inference after Model Averaging in Linear Regression Models," *Econometric Theory*, 35, 816–841.

Supplementary Appendix to "Projected Local Projections"

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A Constrained Local Projections with Piecewise Quadratic Restrictions

We study the estimation of impulse response functions using local projections subject to smoothness constraints. Specifically, we impose that the impulse responses are continuous and continuously differentiable (C^1) piecewise quadratic functions of the forecast horizon.

A.1 Unconstrained Local Projections

Let $\{y_t\}$ be a scalar outcome and x_t a scalar shock observed at time t, with additional control variables $z_t \in \mathbb{R}^m$. For each forecast horizon $h = 0, 1, \ldots, H$, we define the local projection equation:

$$y_{t+h} = \beta(h)x_t + z_t'\gamma(h) + \varepsilon_{t+h}. \tag{A.1}$$

We estimate the impulse response coefficients $\beta(h)$ by first residualizing both y_{t+h} and x_t with respect to z_t using linear projection:

$$\tilde{y}_{t+h} = y_{t+h} - z_t'\hat{\gamma}(h), \qquad \tilde{x}_t = x_t - z_t'\hat{\delta},$$

and then regressing $\tilde{y}_{t+h}$ on $\tilde{x}_t$. Let T be the sample size, and let $Y \in \mathbb{R}^{T(H+1)}$ stack the residualized outcomes $\tilde{y}_{t+h}$ across t and h, and let $\beta \in \mathbb{R}^{H+1}$ collect the impulse responses $\beta(0), \ldots, \beta(H)$. The unconstrained regression system is:

$$Y = X\beta + \varepsilon, \tag{A.2}$$

where $X \in \mathbb{R}^{T(H+1)\times(H+1)}$ is a block-diagonal matrix with blocks of the form $x_t I_{H+1}$:

$$X = \begin{bmatrix} \tilde{x}_1 I_{H+1} \\ \tilde{x}_2 I_{H+1} \\ \vdots \\ \tilde{x}_T I_{H+1} \end{bmatrix}.$$

The OLS estimator is:

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'Y. \tag{A.3}$$

A.2 Piecewise Quadratic Restrictions with Continuity

We impose that the impulse response $\beta(h)$ is continuous and continuously differentiable across a partition of the horizon into K segments,

$$0 = \tau_1 < \tau_2 < \cdots < \tau_{K+1} = H,$$

with the convention that segment k covers the integer grid $\{\tau_k, \ldots, \tau_{k+1}\}$. On segment k we write

$$\beta(h) = a_k + b_k (h - \tau_k) + c_k (h - \tau_k)^2, \qquad h \in \{\tau_k, \ldots, \tau_{k+1}\}. \tag{A.4}$$

Stack $\theta = (a_1, b_1, c_1, \ldots, a_K, b_K, c_K)' \in \mathbb{R}^{3K}$ and define a basis matrix $B \in \mathbb{R}^{(H+1)\times 3K}$ so that $\beta = B\theta$, where each row of B picks the segment and local powers.

Let $\Delta_k = \tau_{k+1} - \tau_k$. For $k = 1, \ldots, K-1$,

$$a_k + b_k\Delta_k + c_k\Delta_k^2 = a_{k+1}, \tag{\textbf{??}}$$

$$b_k + 2c_k\Delta_k = b_{k+1}. \tag{\textbf{??}}$$

Collect them as $C\theta = 0$ with $C \in \mathbb{R}^{(2K-2)\times 3K}$ full row rank.

A.3 Constrained least squares and projection representation

Let Y stack all y_{t+h} and let X map β to $X\beta$ in the stacked regression. The problem is

$$\min_{\theta} \|Y - XB\theta\|^2 \quad \text{s.t.} \quad C\theta = 0.$$

Write $Z := XB$ and assume $Z'Z$ nonsingular. Then

$$\hat{\theta}_{\mathrm{CLS}} = \hat{\theta}_{\mathrm{OLS}} - (Z'Z)^{-1}C'\big(C(Z'Z)^{-1}C'\big)^{-1}C\,\hat{\theta}_{\mathrm{OLS}} =: P_C\,\hat{\theta}_{\mathrm{OLS}},$$

with $\hat{\theta}_{\mathrm{OLS}} = (Z'Z)^{-1}Z'Y$. Here

$$P_C = I - (Z'Z)^{-1}C'\big(C(Z'Z)^{-1}C'\big)^{-1}C$$

is the $(Z'Z)$-orthogonal projector onto $\ker(C)$ (symmetric and idempotent). Then $\hat{\beta}_{\mathrm{CLS}} = BP_C\hat{\theta}_{\mathrm{OLS}}$.

A.4 Linearity and idempotence for horizon–separable scalar regressors

If x_t is scalar and common across horizons, the columns of X are disjoint across h and

$$X'X = \mathrm{diag}(\kappa_0, \ldots, \kappa_H) =: W, \qquad \kappa_h = \sum_{t=1}^{T-h} x_t^2.$$

Thus $Z'Z = B'WB =: G_W$ and $Z'Y = B'W\,\hat{\beta}_{\mathrm{OLS}}$, where $\hat{\beta}_{\mathrm{OLS}} = W^{-1}X'Y$. Substituting gives

$$\hat{\beta}_{\mathrm{CLS}} = P_W\,\hat{\beta}_{\mathrm{OLS}}, \qquad P_W := B\,P_C\,G_W^{-1}B'W.$$

A.5 Proof of idempotence

Let $G_W = B'WB$ and $P_C = I - G_W^{-1}C'(CG_W^{-1}C')^{-1}C$ (symmetric, idempotent). Then

$$P_W^2 = BP_CG_W^{-1}\underbrace{(B'WB)}_{G_W}P_CG_W^{-1}B'W = BP_C^2G_W^{-1}B'W = P_W.$$

Hence P_W is idempotent. When $W = \kappa I$, it reduces to $BP_C(B'B)^{-1}B'$.

Remark. B and C are constructed from $\tau = (\tau_1, \ldots, \tau_{K+1})$; $W = X'X$ depends only on the regressor. Given (τ, W), the map $\hat{\beta}_{\mathrm{OLS}} \mapsto \hat{\beta}_{\mathrm{CLS}}$ is a linear, idempotent transformation.

B Jackknife Model Averaging

The leave-one-out prediction of observation t for model j is given by:

$$\widehat{Y}_{j,(-t)} = X_t P_j \widehat{\beta}_{OLS,(-t)},$$

where $\widehat{\beta}_{OLS,(-t)}$ is the OLS estimate of β with observation t removed. First note, the full model and leave one out estimates of β are related as:

$$\begin{aligned}
\widehat{\beta}_{OLS} &= \frac{\sum_{s=1}^T x_s Y_s}{\sum_{s=1}^T x_s^2} \\
&= \frac{x_t Y_t}{\sum_{s=1}^T x_s^2} + \frac{\sum_{k\neq t}^T x_k^2}{\sum_{s=1}^T x_s^2} \frac{\sum_{k\neq t}^T x_s Y_s}{\sum_{k\neq t}^T x_k^2} \\
&= \frac{x_t Y_t}{\sum_{s=1}^T x_s^2} + \frac{\sum_{k\neq t}^T x_k^2}{\sum_{s=1}^T x_s^2} \widehat{\beta}_{OLS,(-t)} \\
\Rightarrow \widehat{\beta}_{OLS,(-t)} &= \frac{\sum_{s=1}^T x_s^2}{\sum_{k\neq t}^T x_k^2} \left(\widehat{\beta}_{OLS} - \frac{x_t Y_t}{\sum_{s=1}^T x_s^2} \right)
\end{aligned}$$

Thus, we can write the leave-one-out prediction as:

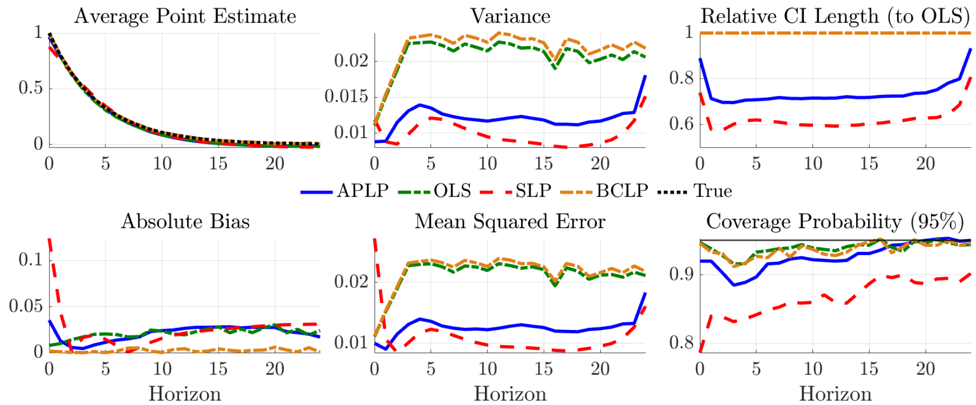
$$\begin{aligned}
\widehat{Y}_{j,(-t)} &= x_t P_j \widehat{\beta}_{OLS,(-t)} \\
&= \frac{\sum_{s=1}^T x_s^2}{\sum_{k\neq t}^T x_k^2} \left(x_t P_j \widehat{\beta}_{OLS} - \frac{x_t^2 P_j Y_t}{\sum_{s=1}^T x_s^2} \right) \\
&= \frac{1}{1-h_t} \left(\widehat{Y}_{j,t} - h_t P_j Y_t \right),
\end{aligned}$$

where $\widehat{Y}_{j,t} = x_t P_j \widehat{\beta}_{OLS}$ is the full-sample prediction and $h_t = x_t^2 / \sum_{s=1}^T x_s^2$ is the leverage of observation t.

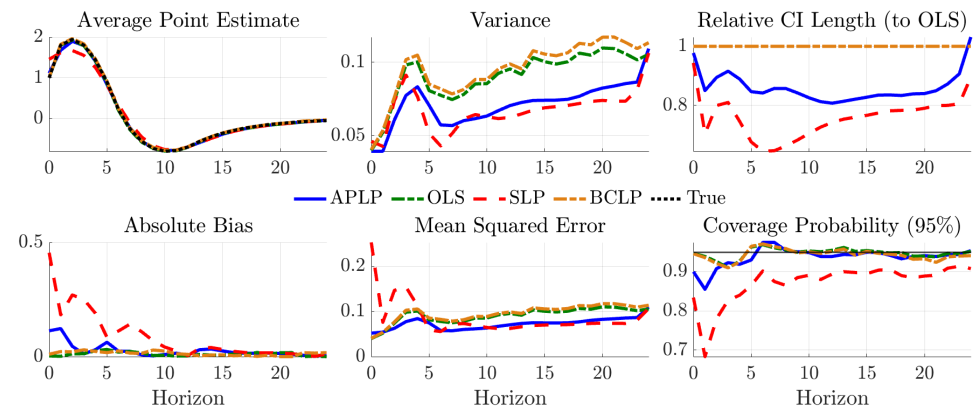
C Additional Simulation Results

Figure A1: Comparison of methods for DGPs with $T = 200$

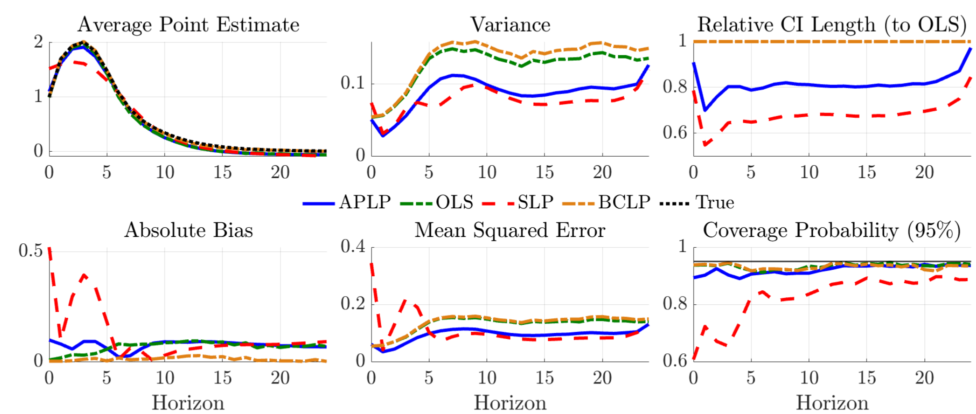
(a) DGP 1: AR(1)



(b) DGP 2: ARMA(1, 12)



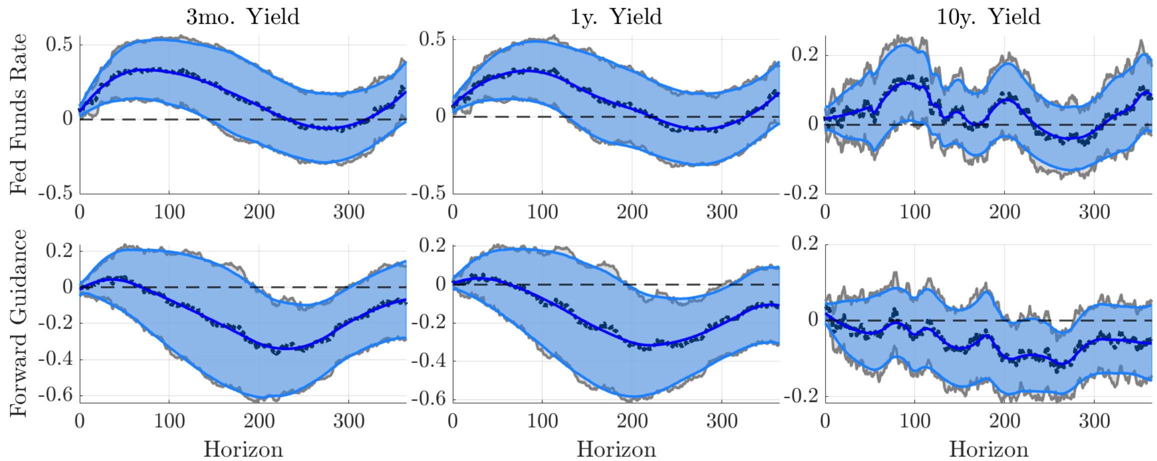
(c) DGP 3: ARMA(3, 5)



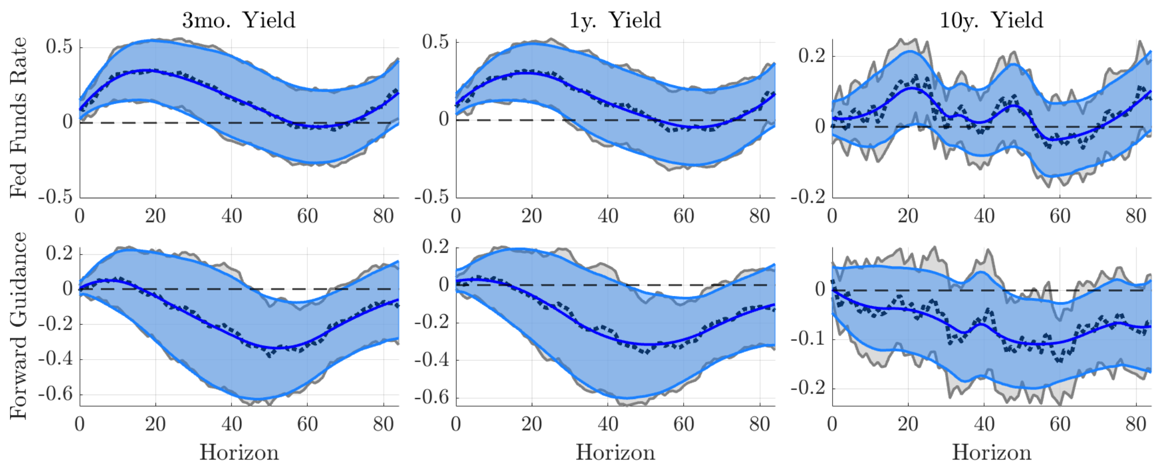
D Additional Figures for Applications

Figure A2: Fed Funds Shocks versus Forward Guidance – Effects on Different Maturities

(a) Weekly Frequency



(b) Monthly Frequency



(c) Quarterly Frequency